# STUDENT ASSIGNMENT & DEADLINE TRACKER

A Mini Project Report submitted to Chaitanya (Deemed to be University)
in partial fulfillment of minimum academic requirements for

**BACHELOR OF TECHNOLOGY**
**COMPUTER SCIENCE AND ENGINEERING**

**Submitted By**


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Under the guidance of

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**
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HYDERABAD – 500075


(2026–2027)

**FACULTY OF ENGINEERING**
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# CERTIFICATE

This is to certify that the Mini Project entitled “Student Assignment & Deadline Tracker” is being submitted by NAME (Roll No), NAME (Roll No), NAME (Roll No), NAME (Roll No), in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at Chaitanya (Deemed to be University) during the academic year 2026–2027.

Name and Signature of Project Guide

(Guide Name and Designation)

Name and Signature of HOD

(Dr. E. Aravind, Assoc. Professor)

Name and Signature of Dean

(Prof. M. Jagadeeshwar)

Signature of External Examiner with Date

# ACKNOWLEDGEMENT

The successful completion of this mini project would not have been possible without the timely help and guidance of many people. We express our sincere gratitude to everyone who supported us throughout the development of this project.

We extend our heartfelt thanks to our project guide, GUIDE NAME (DESIGNATION), Department of Computer Science and Engineering, for valuable guidance, encouragement and suggestions during the planning and implementation of the Student Assignment & Deadline Tracker.

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We thank the faculty members of the Department of Computer Science and Engineering for their support. Finally, we thank our families and friends for their encouragement and support.

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# DECLARATION

We hereby declare that the mini project report entitled “Student Assignment & Deadline Tracker” is an original work carried out by our team at the Faculty of Engineering and Technology, Hyderabad, under the guidance of GUIDE NAME, Department of Computer Science and Engineering, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

We further declare that this report has not been submitted, in whole or in part, to any other institution or university for the award of any degree or qualification. The project work and its documentation have been prepared by our team for academic purposes.

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# ABSTRACT

Student Assignment & Deadline Tracker is a simple web-based application designed to help college students organize assignments and monitor submission deadlines. Students can create assignments with a title, subject, description and due date; view all tasks on a dashboard; update task details; mark tasks as completed; and remove tasks. The system automatically identifies overdue assignments by comparing each due date with the current date. Python and Flask handle the application logic, HTML and CSS provide the interface, and SQLite stores the information locally. The project demonstrates basic web development, database operations and CRUD functionality in a small, explainable prototype.

# 1. INTRODUCTION

Students often receive assignments across multiple subjects and may record deadlines in notebooks, chat messages, calendars or separate files. When details are scattered, a student may overlook a due date or find it difficult to understand which tasks remain unfinished. A centralized assignment tracker provides a simple way to record tasks and check progress.

## 1.1 Problem Statement

College students may struggle to organize assignments and submission deadlines when task information is scattered across different places. This can lead to missed deadlines, forgotten tasks and poor visibility of academic workload. A simple, centralized application is needed to record assignments, display their status and highlight overdue work.

## 1.2 Proposed Solution

The proposed solution is a web application that lets students add, view, edit, complete and delete assignment records. The dashboard displays summary counts for total, pending, completed and overdue tasks. Assignment data is saved in an SQLite database so it remains available after the page is refreshed or the application is restarted.

# 2. OBJECTIVES AND SCOPE

The main objectives are:

- Provide a simple interface for recording assignment details.
- Store assignment information in a database.
- Allow users to edit assignment details and delete unwanted records.
- Track task status as Pending or Completed.
- Identify unfinished assignments whose due dates have passed.
- Demonstrate the fundamentals of a web application using beginner-friendly technologies.

## Scope and limitations

This prototype is intended for local demonstration by a student or a small team. It does not include user accounts, multi-user access, email reminders, file uploads or cloud synchronization. These features can be considered future enhancements if required by the project guide.

# 3. TECHNOLOGIES USED

| Technology | Purpose in the project |
|---|---|
| Python | Implements the application logic and date/status calculations. |
| Flask | Connects browser requests and form submissions to Python functions. |
| HTML | Defines the forms, dashboard, assignment list and page structure. |
| CSS | Styles the layout, cards, buttons, task status labels and responsive design. |
| SQLite | Stores assignment records in a local database file. |

# 4. SYSTEM DESIGN

## 4.1 Main modules

• Assignment entry: validates and saves a new assignment. • Assignment dashboard: retrieves records and displays summary counts. • Status management: toggles a task between Pending and Completed. • Edit and delete: updates an existing record or removes it. • Overdue detection: flags a task when its due date is earlier than today and its status is not Completed.

## 4.2 Data model

The assignments table contains the following fields: id (unique primary key), title, subject, description, due_date and status. The id is generated automatically. Title, subject and due_date are required; description is optional. Status is restricted to Pending or Completed.

## 4.3 Application flow

1. The user opens the home page in a browser. 2. Flask queries SQLite for assignment records. 3. Python calculates the total, pending, completed and overdue counts. 4. Flask sends the data to the HTML template. 5. When the user submits a form, Flask validates the values and performs the corresponding database operation. 6. The browser returns to the dashboard and displays the updated information.

# 5. IMPLEMENTATION

The application is implemented in app.py. Flask routes handle the home page, adding an assignment, toggling status, editing and deleting. Database operations use parameterized SQL queries. The interface is defined in templates/index.html and its styling is defined in static/style.css. On the first run, the application creates the assignments table if it does not already exist.

## 5.1 How overdue status works

The application obtains the current date and compares it with each assignment's due_date. If the due date is earlier than today and the status is not Completed, the assignment is labelled Overdue. The overdue value is calculated when the dashboard loads rather than being permanently stored in the database.

## 5.2 How to run the prototype

Install Python 3.9 or later, open a terminal in the project folder, install the dependency with “pip install -r requirements.txt”, and run “python app.py”. Then open http://127.0.0.1:5000 in a browser. The database file is created automatically.

# 6. TESTING

| Test case | Expected result |
|---|---|
| Add a valid assignment | Record is saved and appears in the assignment list. |
| Submit without title/subject/date | Validation message appears and no record is added. |
| Mark pending task completed | Status changes to Completed and summary counts update. |
| Use a past due date for a pending task | Task receives the Overdue label. |
| Edit an existing assignment | Updated details appear after returning to the dashboard. |
| Delete an assignment | Record is removed from the list. |
| Refresh or restart the app | Saved records remain in the SQLite database. |

These are the planned functional checks for the supplied prototype. The team should execute each test on its own machine before the review and record actual screenshots/results in the final submission.

# 7. CONCLUSION AND FUTURE ENHANCEMENTS

The Student Assignment & Deadline Tracker provides a straightforward way to manage academic tasks in one place. It demonstrates CRUD operations, form validation, database persistence, status tracking and date-based logic using Python, Flask, HTML, CSS and SQLite. The project is intentionally small so the team can explain the flow clearly during a viva and demonstrate its main features.

Future enhancements may include student login, separate user dashboards, search and filtering, calendar view, reminder notifications, assignment file attachments and deployment to a web server. These are not included in the current prototype.

# REFERENCES

1. Python Software Foundation, Python Documentation: https://docs.python.org/3/

2. Pallets Projects, Flask Documentation: https://flask.palletsprojects.com/

3. SQLite, SQLite Documentation: https://www.sqlite.org/docs.html

4. MDN Web Docs, HTML and CSS guides: https://developer.mozilla.org/

# APPENDIX: TEAM COMPLETION CHECKLIST

Before printing and submitting, replace every placeholder with the actual team member names, roll numbers and guide details; run the prototype on the team's laptop; capture screenshots of adding, completing, editing and deleting assignments; verify the overdue label using a past date; and confirm the final report requirements with the project guide.